CS185 Homework 5 Report

Offline RL

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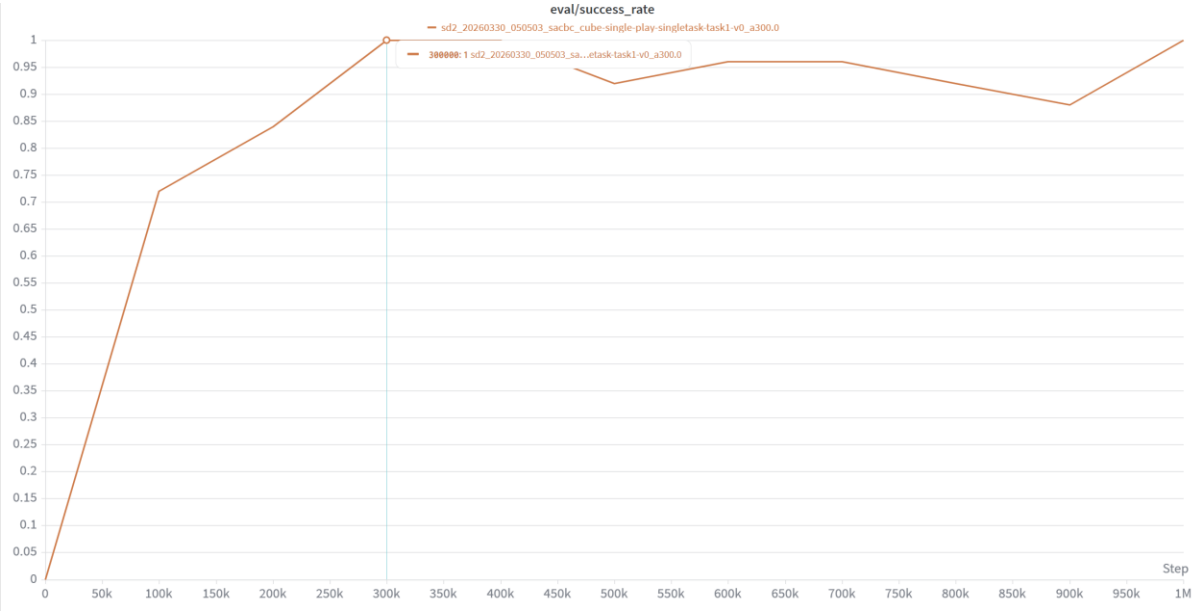
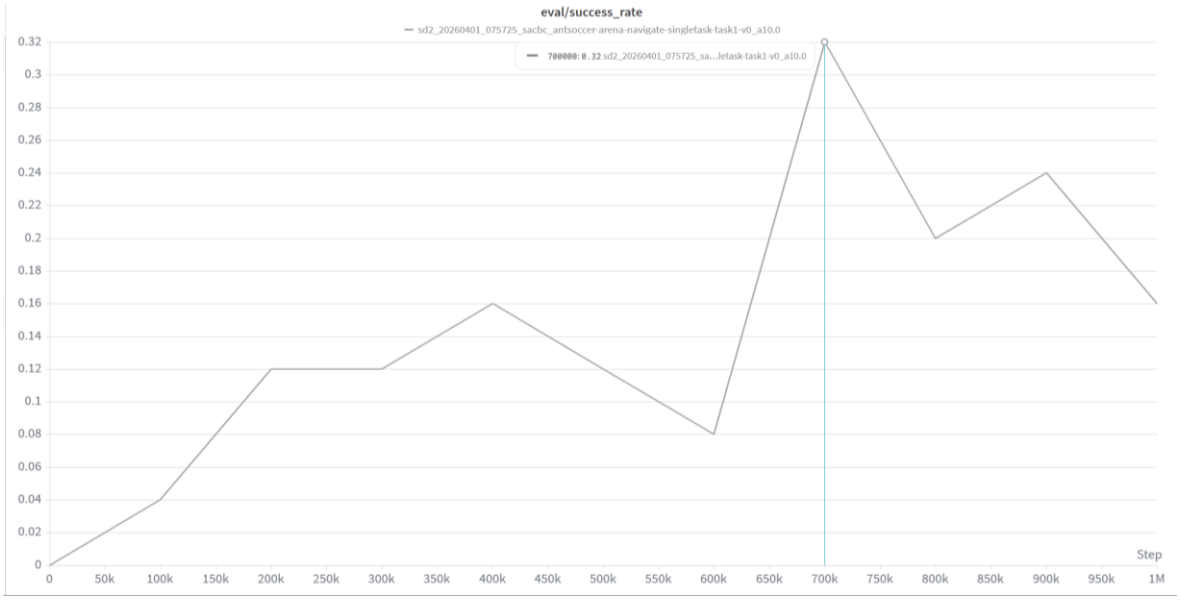
2. Offline RL with Behavioral Regularization

– Training curves (training steps vs. success rate) of your best SAC+BC agents on the above two tasks.

cube-single-play-singletask-task1-v0

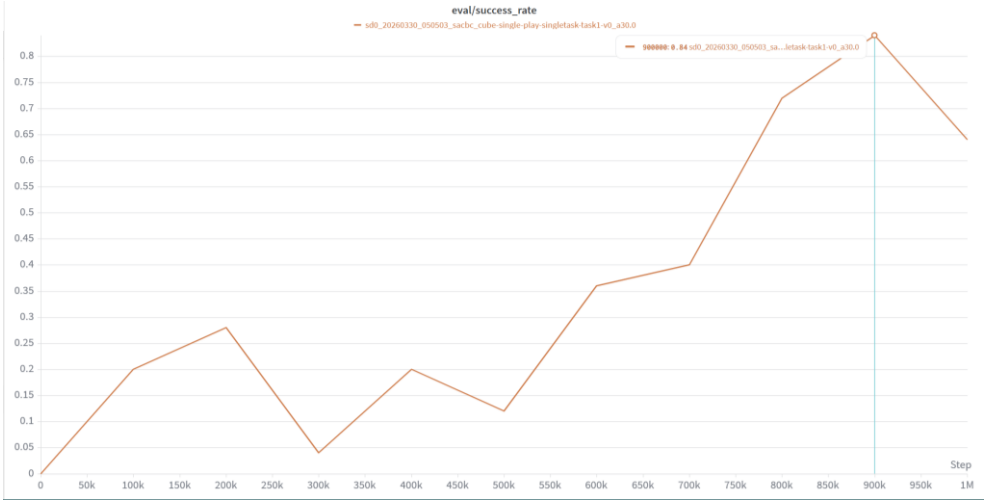


antsoccer-arena-navigate-singletask-task1-v0

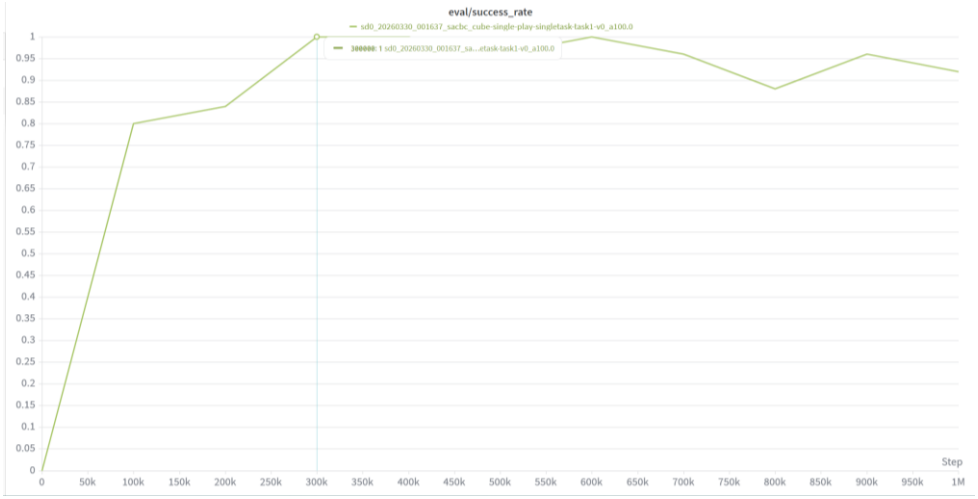


– Training curves with different values of α (of your choice, at least 3 different values) on cube-single.

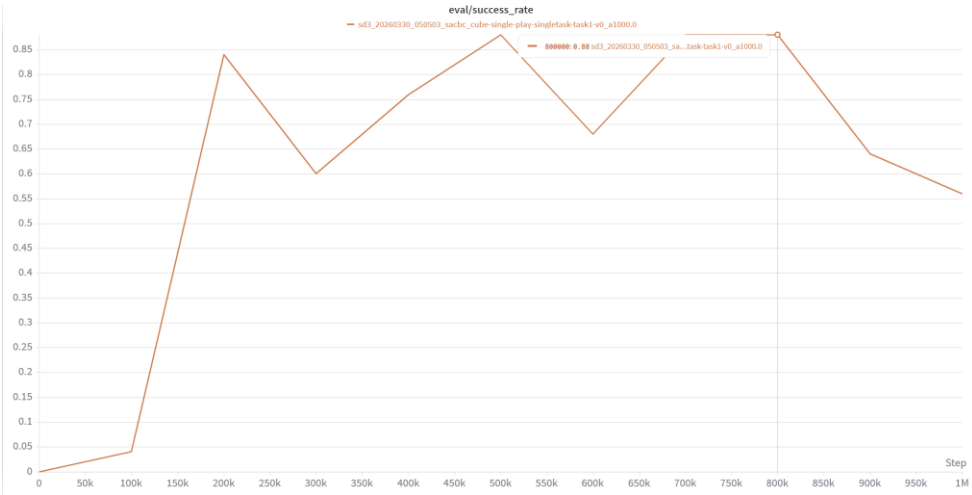
α = 30.0



α = 100.0



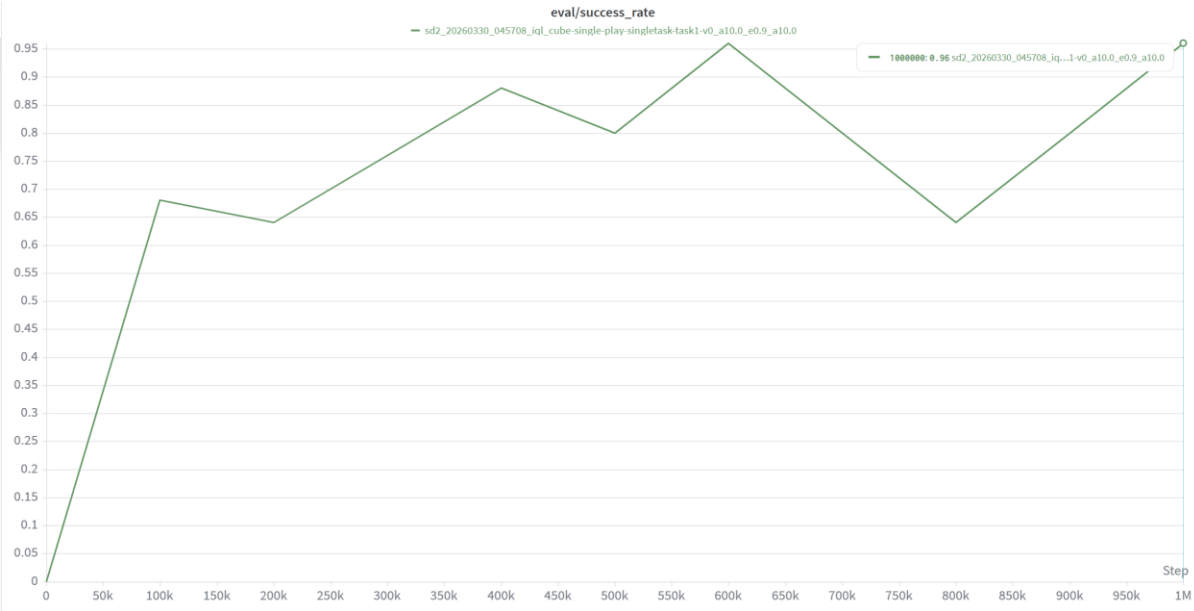
α = 1000.0



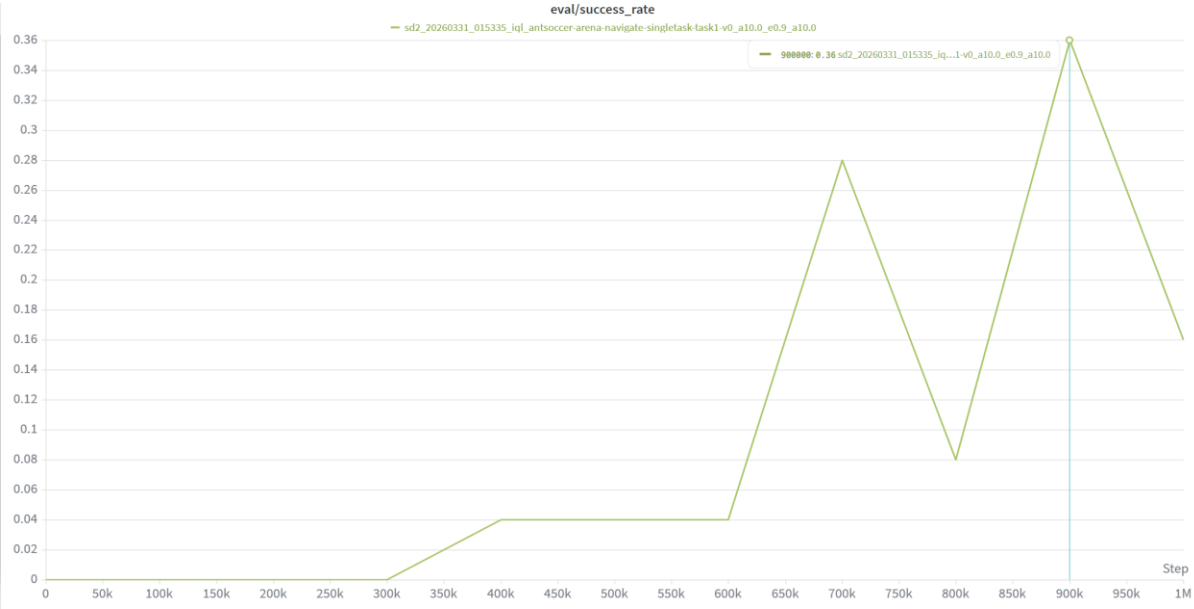
3. Offline RL with Implicit Q-Learning

– Training curves (training steps vs. success rate) of your best IQL agents on the above two tasks.

cube-single-play-singletask-task1-v0

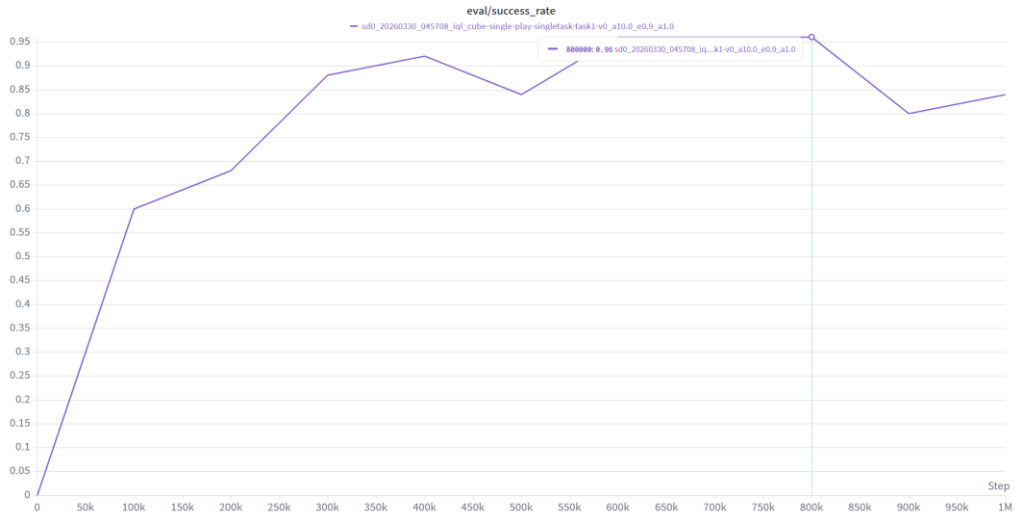


antsoccer-arena-navigate-singletask-task1-v0

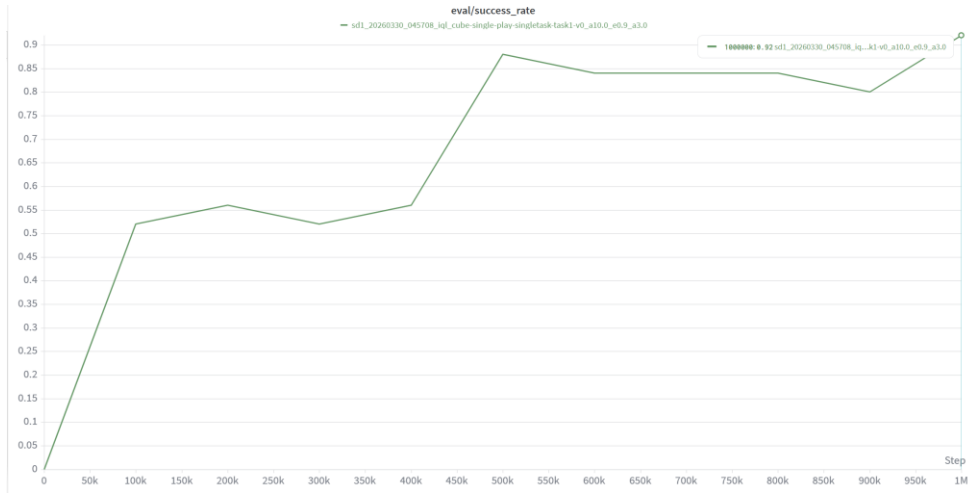


– Training curves with different values of α (of your choice, at least 3 different values) on cube-single.

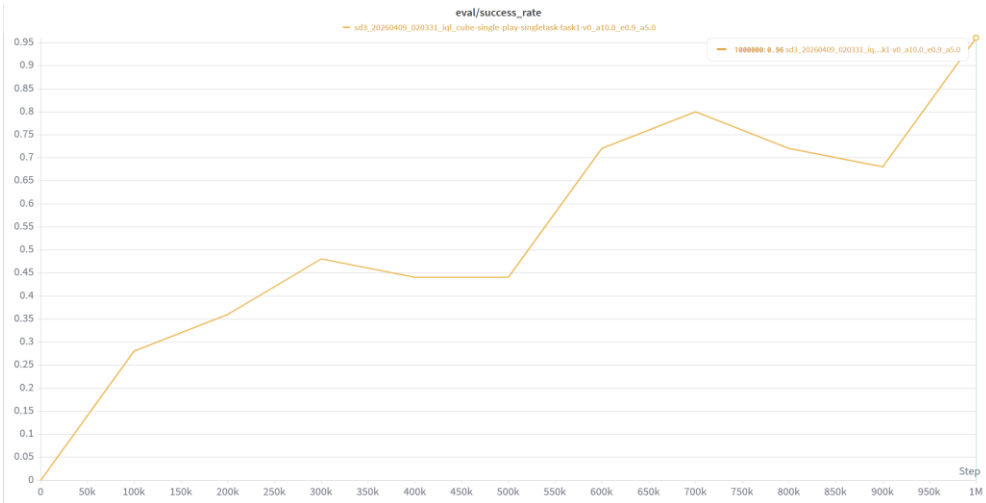
α = 1.0



α = 3.0



α = 5.0



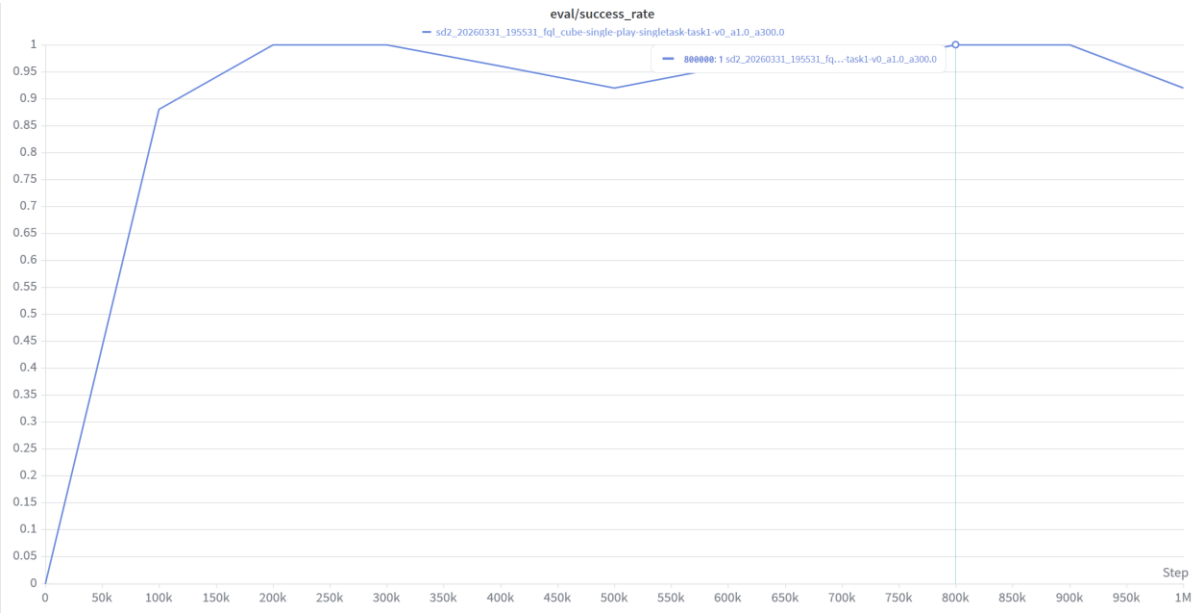
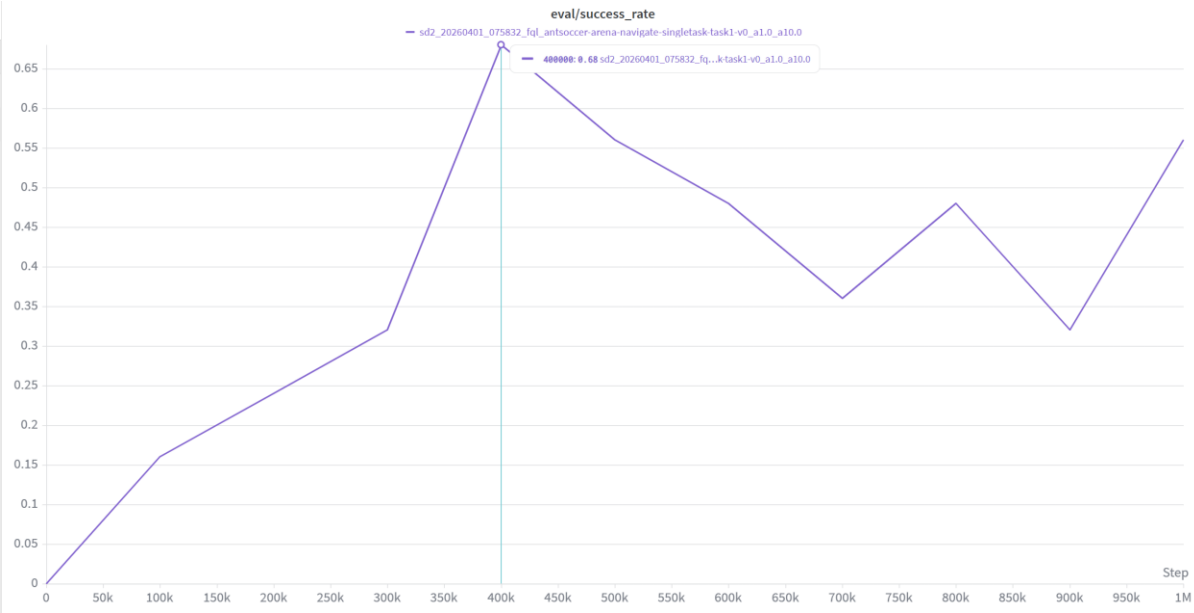
– A short discussion (1–2 sentences) that compares SAC+BC and IQL in terms of (1) the best performance and (2) sensitivity to α.

SAC+BC performs better in optimal conditions for cube-single task but IQL performs better in optimal conditions on the antsoccer task. SAC+BC is more sensitive to α than IQL.

4. Offline RL with Flow Policies

- Training curves (training steps vs. success rate) of your best FQL agents on the above two tasks.

cube-single-play-singletask-task1-v0



antsoccer-arena-navigate-singletask-task1-v0